Yan (Annie) Zhu

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EDUCATION

University of Michigan–Ann Arbor, *College of Engineering* **Ann Arbor, MI**
Bachelor of Science in Engineering *in Robotics Engineering* *Expected May 2028*

- *Courses:* SLAM & Navigation; Programming & Data Structures, Computational Symmetry in AI and Robotics

Northeastern University, *College of Engineering* **Boston, MA**
Bachelor of Science *coursework in Industrial Engineering* *Sept 2024 – May 2026*

- *Academics:* **GPA 3.84/4.0** | Presidential Global Scholarship ($9,000), John Martinson Honors Program, Dean's List

TECHNICAL SKILLS

CAD & Prototyping: SolidWorks, Rhino, AutoCAD, 3D printing, laser cutting, soldering, woodworking
Programming & Robotics: C++, Python, ROS 2, MATLAB, SLAM, Monte Carlo localization, Rviz
Lab Skills: Resistance measurement, optical microscopy, thermal evaporation, spin coating
Systems & Documentation: Visio, network diagrams, requirements analysis, technical documentation

PROJECT EXPERIENCE

MBot SLAM & Autonomous Navigation **Ann Arbor, MI**
ROB 330 Team Project, University of Michigan *Sept 2026 – Present*

- Tuned PID controllers for wheel-speed tracking and differential-drive motion; performed UMBmark calibration using clockwise and counterclockwise square-path tests to evaluate systematic errors with wheel diameter and wheelbase
- Integrated occupancy-grid mapping with particle-filter localization using action and sensor models in C++/ROS 2; debugged SLAM map updates and LiDAR alignment using recorded data and physical-maze tests
- Implemented and tested A* and Theta* path planners for map-based navigation; worked on waypoint-control integration and diagnosed ROS 2 transform and initial-pose alignment issues using Rviz

Multi-Module MAG-LEV Train System **Oakland, CA**
Team Lead & Product Developer *Jan 2025 – May 2025*

- Led a team of 5 in developing a small-scale, modular magnetic-levitation model train for third-grade students, combining a self-assembled mechanical structure with elastic energy storage and motor control
- Designed 81 components in SolidWorks and Rhino and fabricated the model using 3D printing, laser cutting, and woodworking, carrying the design from digital component models through physical prototype assembly
- Conducted 3 on-site user tests at Mills College Children's School and gathered feedback to guide usability improvements; the team's model train design received the most favorable feedback from student users

RESEARCH EXPERIENCE

Gong Research Group, University of Michigan **Ann Arbor, MI**
Undergraduate Researcher, Wearable Optoelectronics *Oct 2026 – Present*

- Supported experimental design and nanoscale film fabrication for mechanically robust optoelectronics, using thermal evaporation and spin/spray coating to prepare samples and electrodes for characterization
- Measured electrical resistance and performed optical characterization to assess fabricated samples; used optical microscopy to inspect samples during electrode preparation and testing

INTERNSHIP EXPERIENCE

DNV Class Society **Shanghai, China**
Ship Cybersecurity Consulting Intern, Supply Chain & Product Assurance *Jun 2025 – Aug 2025*

- Built a reference database consolidating device and security requirements from 350+ pages across 8 maritime regulatory frameworks, reducing engineers' compliance review time from approximately 3 hours to 40 minutes
- Produced a 40-page usage guide interpreting technical requirements and authored 4 compliance articles with 3,500+ views, making regulatory guidance accessible to engineers and shipowners

AquaSync Innovation (ASI) **Singapore & Shanghai, China**
Ship Cybersecurity Analyst Intern, Marine IT/OT Systems *Apr 2024 – Aug 2024*

- Created 4 network topology diagrams in AutoCAD and Visio from ship asset inventories; designed and presented firewall solutions based on device configurations, system constraints, and perimeter protection principles
- Supported development of a unit integrating firewall, IDS, server, and switch functions; coordinated requirements across 10 manufacturers and evaluated production constraints through a bidding process that shortlisted 1 finalist
- Conducted an onboard vulnerability scan of a 114,400-DWT oil tanker and prepared a 45-page risk assessment report for regulatory review, documenting findings for technical stakeholders